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Risk-averse stochastic optimization

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- Risk measures
- Risk-averse stochastic optimization



- From Nicolas Bernoulli's letter 1713
- **St. Petersburg game:** flip a fair coin until it comes up heads the 1st time. At that point the player wins 2^n €, where n is the number of times the coin was flipped. How much should one pay for playing this game?

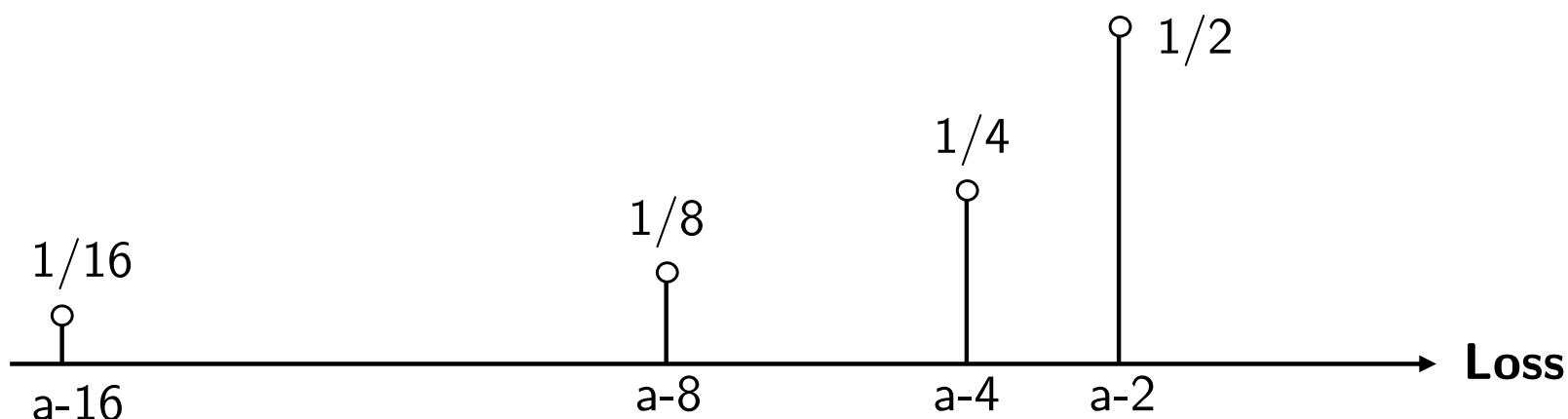
Trial where first tails appears	Probability	Payout
1	$1/2$	2
2	$1/4$	4
3	$1/8$	8
4	$1/16$	16
$\vdots$	$\vdots$	$\vdots$

- Decision theorists' advice: **the expected value**

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} P(X = n) \cdot \text{Payout}(n) = \sum_{n=1}^{\infty} \left(\frac{1}{2^n} \times 2^n \right) = \sum_{n=1}^{\infty} 1 = \infty$$



- The “paradox”: it would be rational to pay any **finite** fee to play the game?
- Proposed resolutions, e.g., Daniel Bernoulli’s **expected utility**
 - $\text{utility} = \log_{10}(2^n)$ (diminishing marginal utility)
 - expected utility ≈ 4
 - satisfactory?
- Why we have the paradox?
 - **Expectation neglects the risk of the (bad) outcomes**





- **Different attitudes** towards the risk of the outcomes

Game A:

20% +1000€, 80% -100 €

Game B:

100% +120€

- **Risk averse** fear loss and seek sureness
- **Risk neutral** are indifferent to the degree of risky outcome
- **Risk seeking** hope to “win big” and don’t mind losing as much



Risk:

The combination of the **uncertainty of occurrence** of “bad outcome”, and **the severity of that outcome**

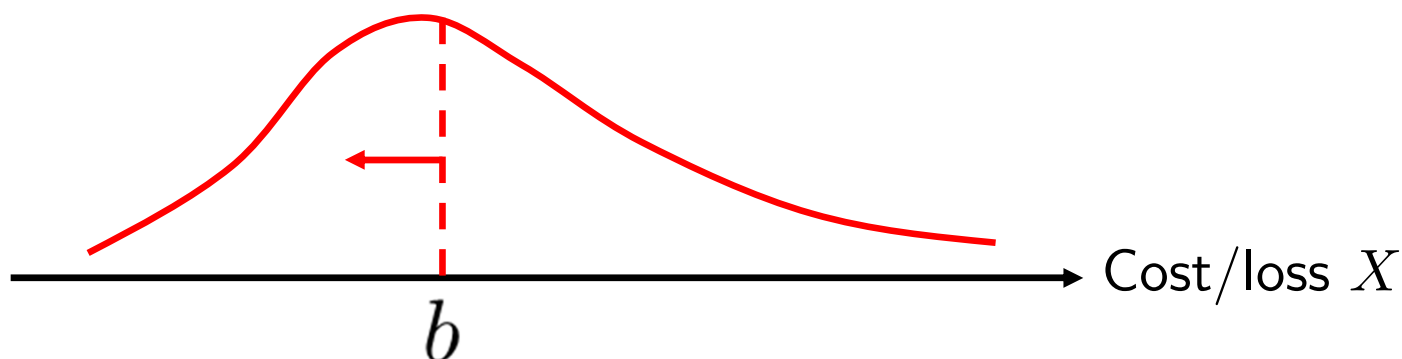
- **Random outcomes:** r.v. $X : \Omega \rightarrow \mathbb{R}$
→ **“Cost/loss” oriented** X : high outcomes bad, low outcomes good
- **Risk measure:** a quantification be applied to X that elicits the level of “cost/loss” in X

$\mathcal{R}(X)$ maps X into $\mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$

- Risk $\neq$ Uncertainty (statistical dispersion)



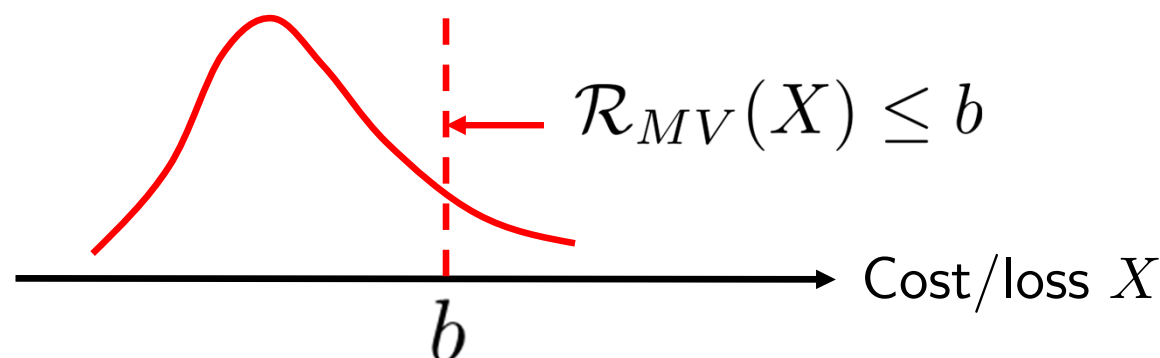
- **Expectation:** $\mathcal{R}(X) = \mathbb{E}[X]$
 - $\mathcal{R}(X) \leq b \Leftrightarrow X \leq b$ on average
 - but not risk averse (risk measure?), perhaps too feeble



- **Focusing on worst cases:** $\mathcal{R}(X) = \sup X$
 - $\mathcal{R}(X) \leq b \Leftrightarrow X \leq b$ almost surely
 - Averse, perhaps overly conservative, infeasible



- **Mean variance:** $\mathcal{R}_{MV}(X) = \mathbb{E}[X] + \lambda \mathbb{V}[X]$



- Widely used in finance (portfolio optimization, Harry Markowitz 1952)
- Possible drawback: variance is **symmetric**! penalizes high cost as well as low cost (profit?)



Value-at-Risk (VaR), α -quantile



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- Loss/cost r.v. X associated with CDF $F(x)$
- For any $\alpha \in [0, 1]$, the VaR on α is defined by

$$\mathbf{VaR}_\alpha(X) = q_\alpha(X) = \inf \{x \in \mathbb{R} : F(x) \geq \alpha\}$$

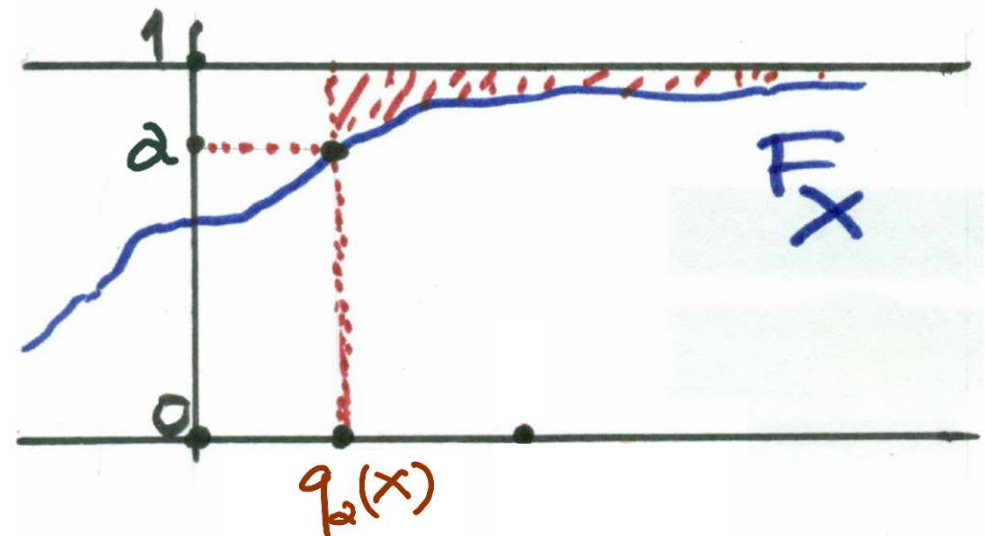
- For **continuous** CDF, $F(\mathbf{VaR}_\alpha) = \alpha$
- For **strictly increasing and continuous** CDF

$$\mathbf{VaR}_\alpha = F^{-1}(\alpha)$$

- Then

$$\mathcal{R}(X) = \mathbf{VaR}_\alpha(X) \leq b$$

$$\Leftrightarrow \mathbb{P}(X \leq b) \geq \alpha$$



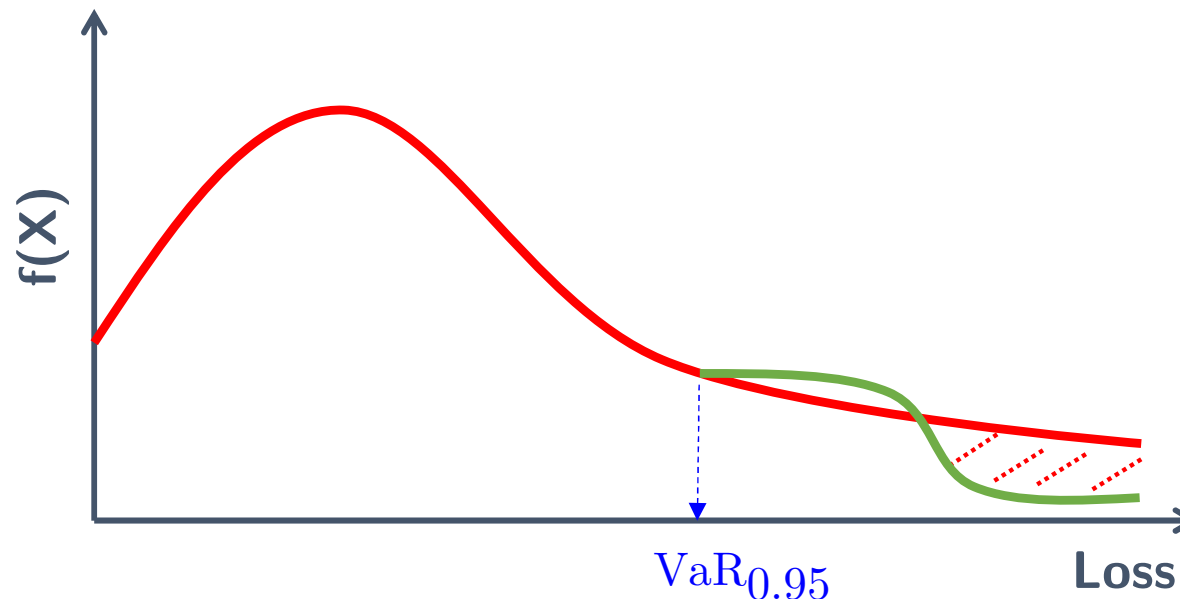


Value-at-Risk (VaR), α -quantile



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- Widely used (in Finance)
 - VaR_α pays no attention to the **magnitude of losses** when the rare event of experiencing a loss above the level VaR_α occurs
 - Bad mathematical & computational behavior: nonconvex, nonlinear
- J.P. Morgan 95%
 - Bank of America 95%
 - Citibank 95.4%
 - Chase Manhattan 97.5%
 - Basel Committee on Bank Supervision 99%



- For $\alpha \in [0, 1]$, the CVaR on α is defined by

$$\mathbf{CVaR}_\alpha(X) = Q_\alpha(X) = \mathbb{E}[X | X \geq \mathbf{VaR}_\alpha(X)]$$

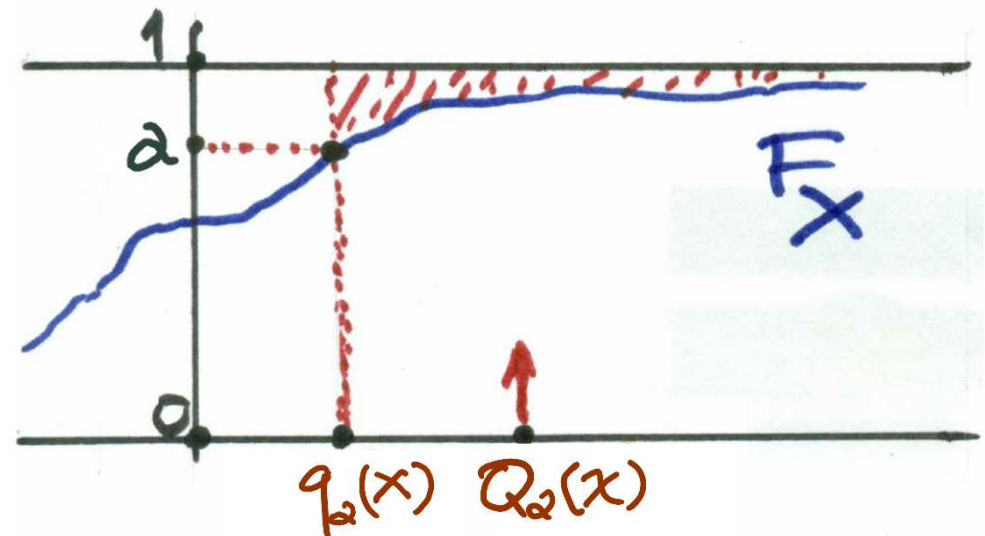
- Or, equivalently

$$\mathbf{CVaR}_\alpha(X) = \frac{1}{1 - \alpha} \int_{x \geq \mathbf{VaR}_\alpha} x dF(x)$$

- Good mathematical behaviors
- Measure the

$$\mathcal{R}(X) = Q_\alpha(X) \leq b$$

$\Leftrightarrow X \leq b$ on average in
upper α -tail





Definition

A risk measure $\mathcal{R}(\cdot)$ is called **coherent** if it satisfies the following properties:

(A1) **Convexity**. $\beta \in (0, 1)$, X_1 and X_2 random variables $\Rightarrow$

$$\mathcal{R}(\beta X_1 + (1 - \beta)X_2) \leq \beta \mathcal{R}(X_1) + (1 - \beta)\mathcal{R}(X_2)$$

(A2) **Monotonicity**. $X_1 \geq X_2$ a.s. $\Rightarrow \mathcal{R}(X_1) \geq \mathcal{R}(X_2)$

(A3) **Translation invariance**. If X is a r.v. and $a \in \mathbb{R}$, then

$$\mathcal{R}(X + a) = \mathcal{R}(X) + a$$

(A4) **Positive homogeneity**. $\mathcal{R}(tX) = t\mathcal{R}(X), \forall t > 0$

References: Artzner et al. (1999), Ruszczynski and Shapiro (2006)



$$\text{VaR}_\alpha(X) = \inf \{x \in \mathbf{R} : F(x) \geq \alpha\}$$

- Convex? **No**
- Monotone? **Yes**
- Translation invariant? **Yes**
- Positively homogeneous? **Yes, if $\mathbf{P}(X \geq 0) = 1$**

Mean variance $\mathcal{R}_{MV}(X) = \mathbb{E}[X] + \lambda \mathbb{V}[X]$

- Convex? **Yes**
- Monotone? **No**
- Translation invariant? **Yes**
- Positively homogeneous? **No**



- Suppose there are **three equally likely outcomes** (loss)

ω	Z^1	Z^2	$\frac{1}{2}Z^1 + \frac{1}{2}Z^2$
1	300	0	150
2	0	0	0
3	0	300	150
VaR _{0.6}	0	0	150
$\mathbb{E}$	100	100	100

- $0.5\mathbf{VaR}_{0.6}(Z^1) + 0.5\mathbf{VaR}_{0.6}(Z^2) < \mathbf{VaR}_{0.6}(0.5Z^1 + 0.5Z^2)$



- Suppose there are **two outcomes**: $P(1) = 0.95$, $P(2) = 0.05$

ω	Z^1	Z^2
1	10	10
2	10	0
$\mathbb{E}$	10	9.5
$\mathbb{V}$	0	4.75
$\mathbb{V}^{1/2}$	0	2.18

- $Z^1 \geq Z^2$ with probability 1, but **mean variance** (or **mean standard-deviation**) would prefer Z^1 for modest values of λ

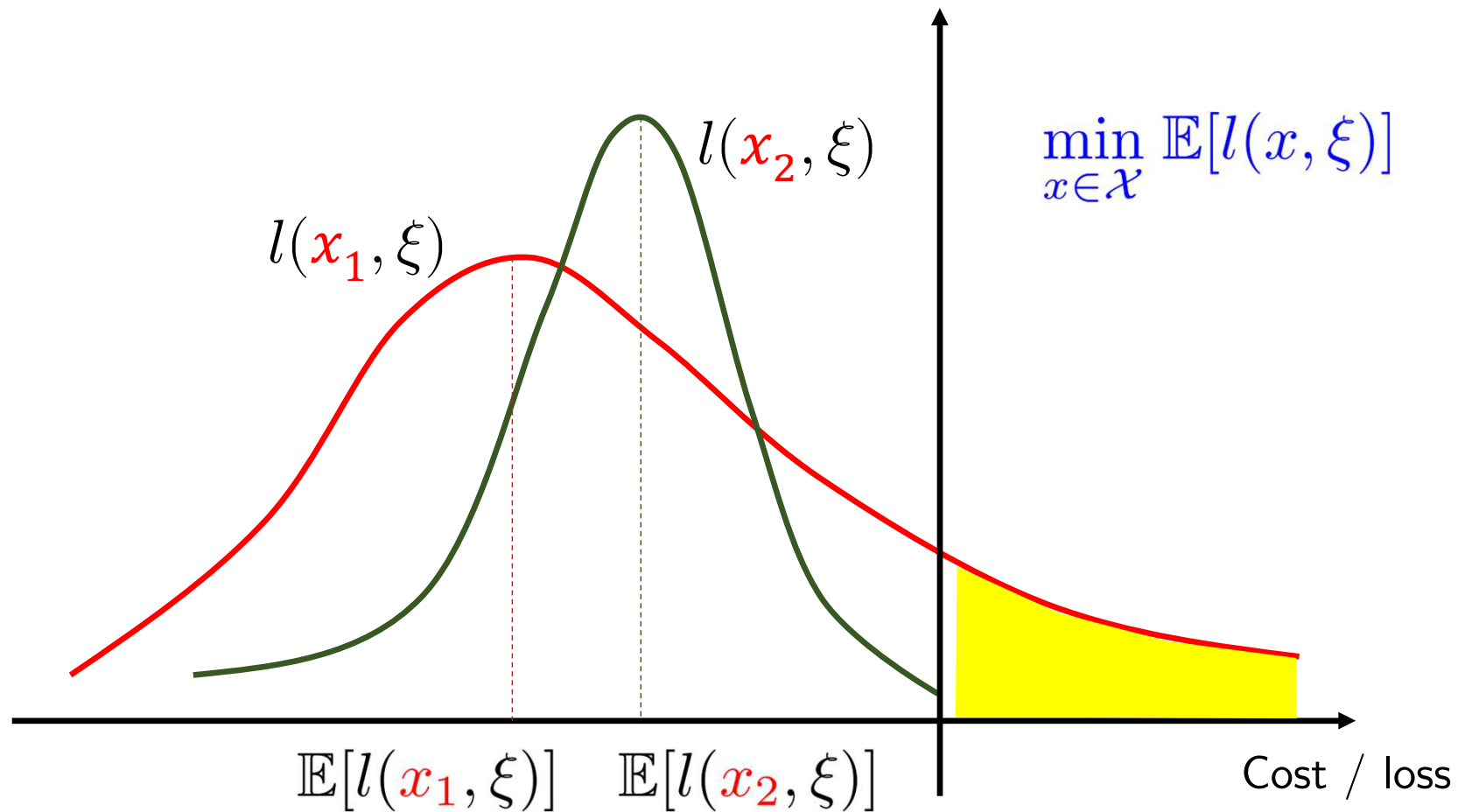


CVaR($\cdot$) is **coherent** (Rockafellar and Uryasev, 2000)

- Convex? **Yes**
- Monotone? **Yes**
- Translation invariant? **Yes**
- Positively homogeneous? **Yes**



- To control of the **tail risk** in risk averse situations





- Classical (Markowitz) portfolio optimization

$$\max_w \quad w^T \mu - \lambda w^T \Sigma w$$

$$\text{s.t.} \quad \sum_{i=1}^n w_i = 1, \quad (\text{Fully invested portfolio})$$

$$w_i \geq 0, \quad \forall i \quad (\text{No short-selling, optional})$$

→ n be the number of assets

→ $w \in \mathbb{R}^n$ be the **decision vector** of asset weights

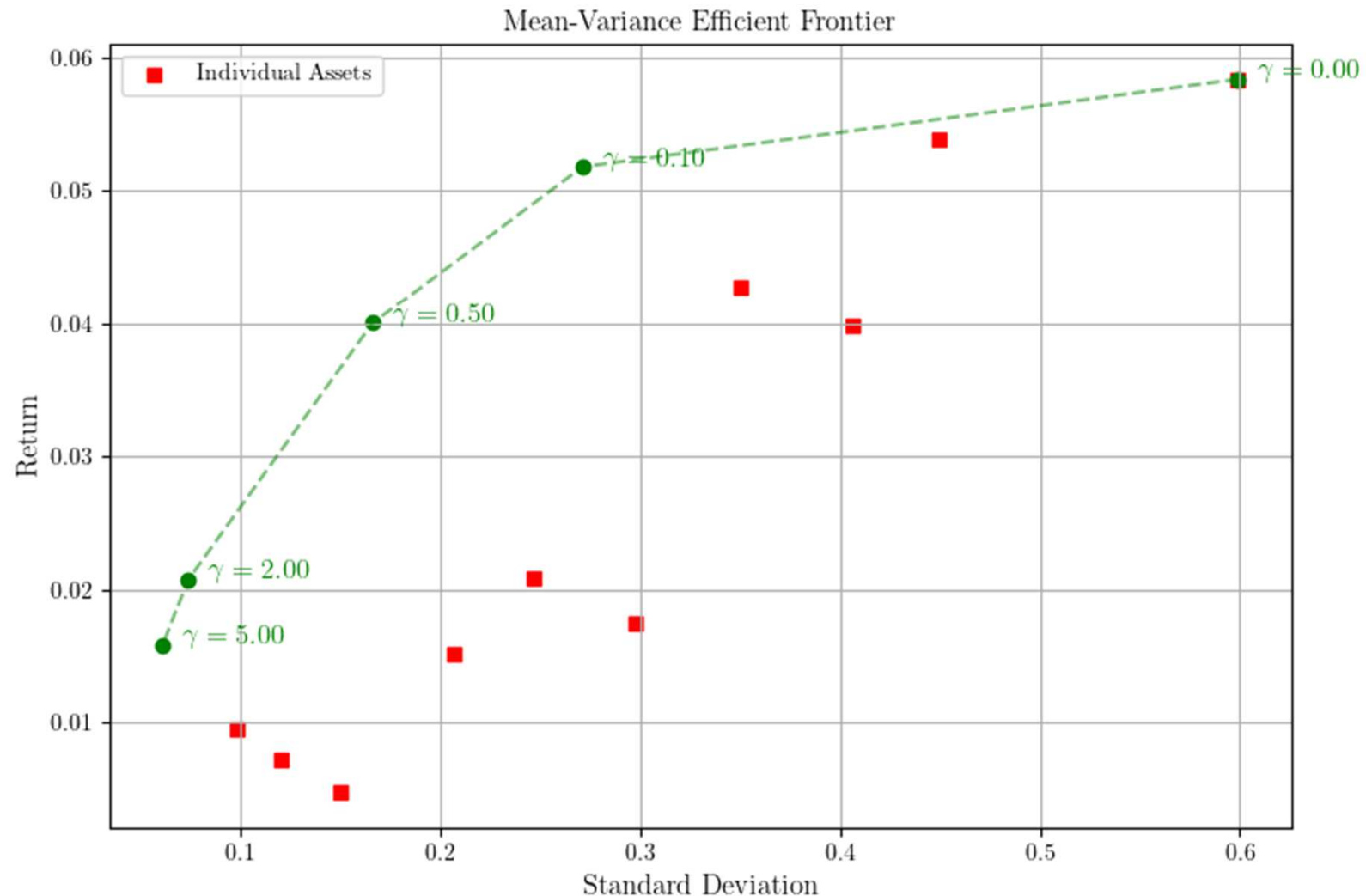
→ $\mu \in \mathbb{R}^n$ be the expected return vector (**parameters**)

→ $\Sigma \in \mathbb{R}^{n \times n}$ be the covariance matrix of returns (**parameters**)

→ $\lambda \geq 0$ be the risk-aversion parameter

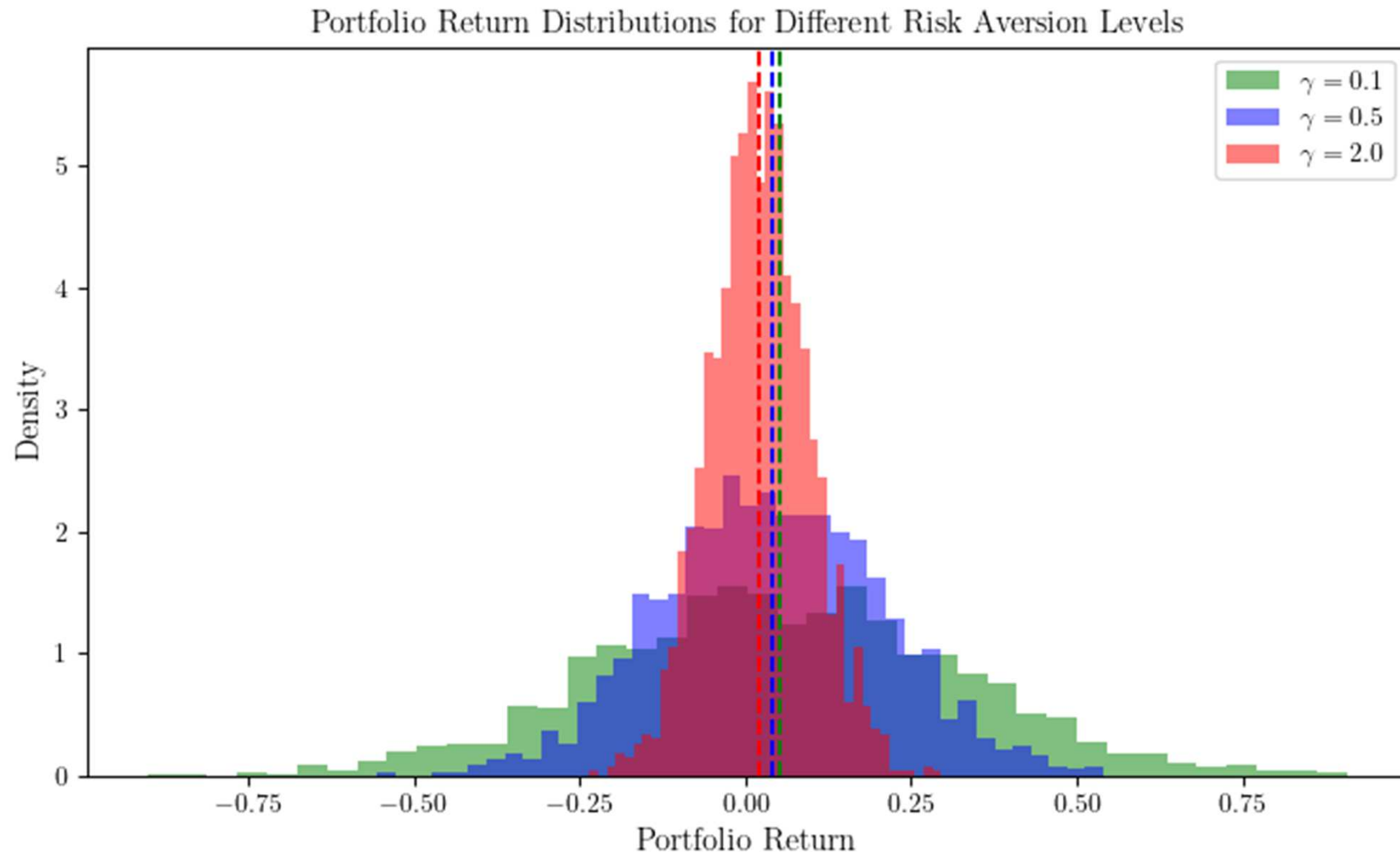


- Optimal risk-return trade-off for 10 assets, long only portfolio





- Return distributions for different risk aversion values





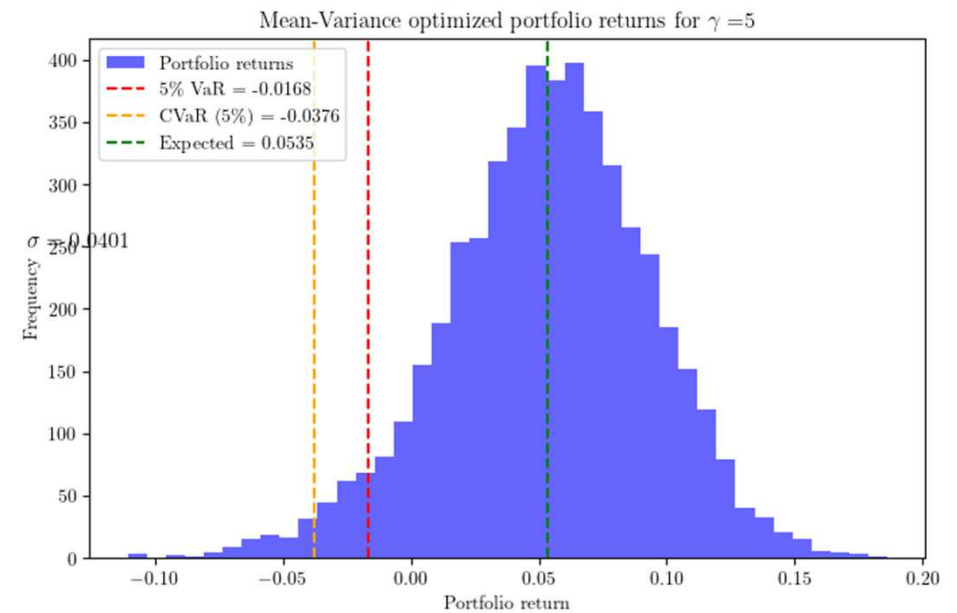
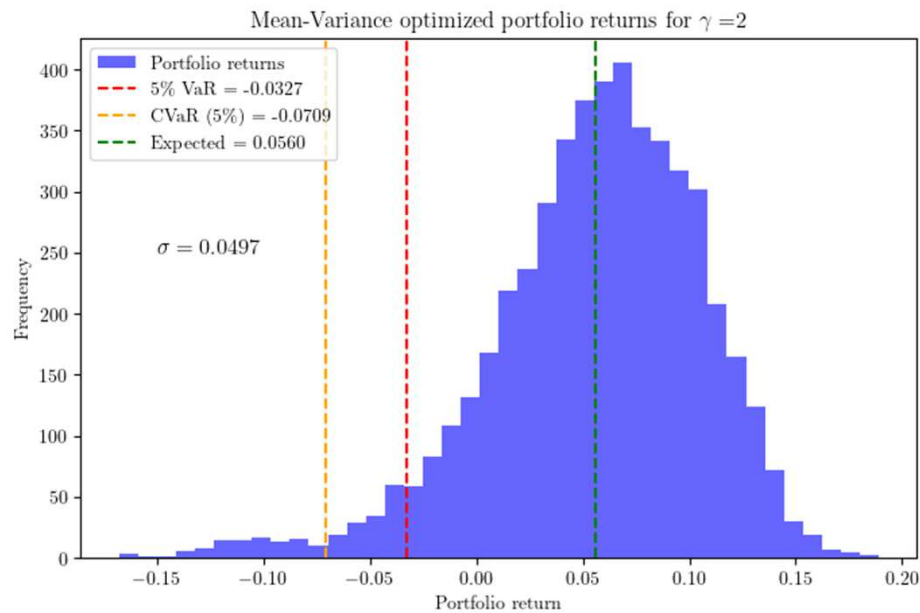
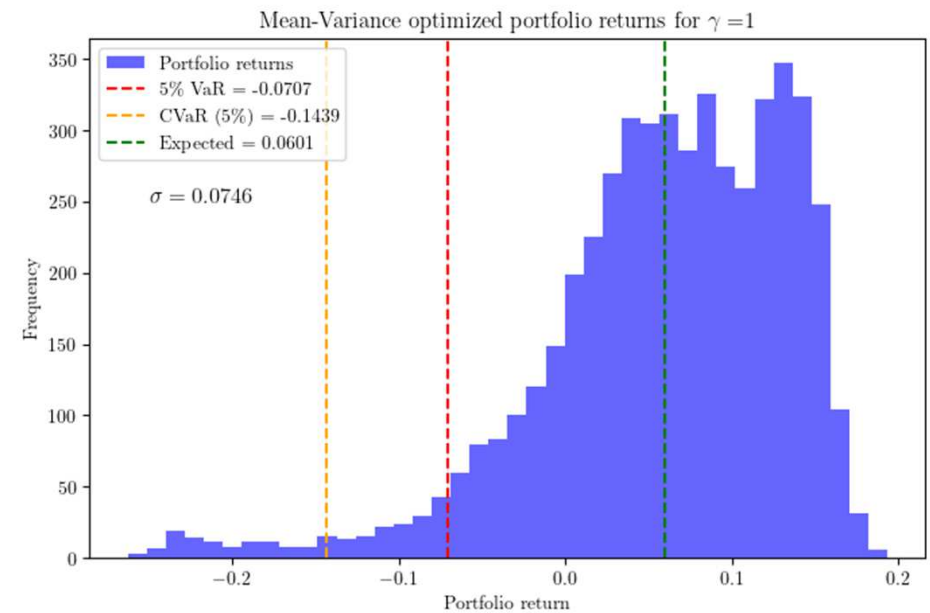
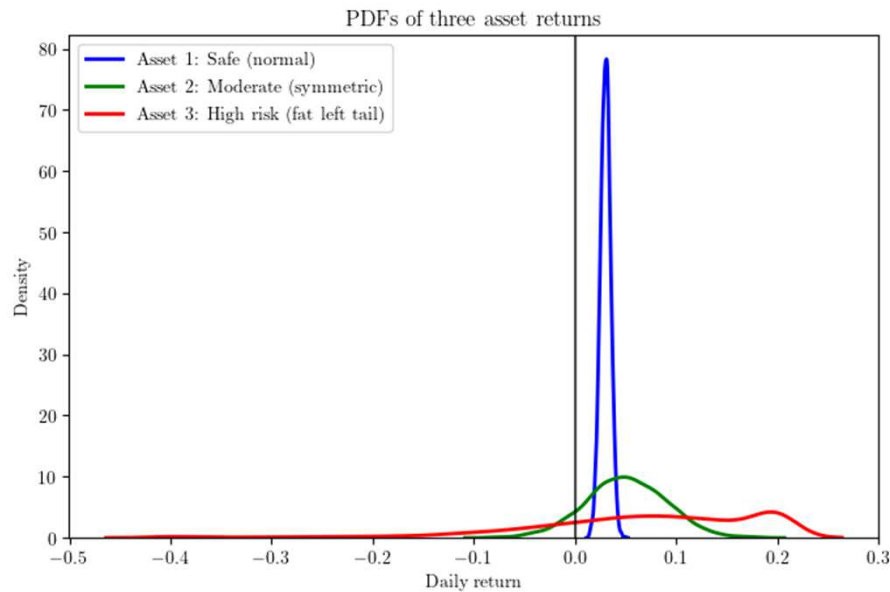
- Penalizes upside volatility
- **No direct control** over tail risk
- Work well in normal distributions, **but poorly in skewed or heavy-tailed** distributions



Mean-variance models



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- Let $l(x, \xi)$ represent a random loss function, by definition:

$$\text{VaR}_\alpha[l(x, \xi)] = \inf \{ \gamma | \mathbb{P}[l(x, \xi) \leq \gamma] \geq \alpha \}$$

- Chance constraint:**

$$\text{VaR}_\alpha[l(x, \xi)] \leq b \Leftrightarrow \mathbb{P}[l(x, \xi) \leq b] \geq \alpha$$

- VaR in the **objective function**

$$\begin{aligned} \min_x \text{VaR}_\alpha[l(x, \xi)] &\Leftrightarrow \min_{x, \gamma} \gamma \\ &\text{s.t. } \mathbb{P}[l(x, \xi) \leq \gamma] \geq \alpha \end{aligned}$$



- For $\alpha \in [0, 1]$, the CVaR on α is defined by

$$\mathbf{CVaR}_\alpha(X) = \mathbb{E}[X | X \geq \mathbf{VaR}_\alpha(X)] = \frac{1}{1 - \alpha} \int_{x \geq \mathbf{VaR}_\alpha} x dF(x)$$

- Computing CVaR generally requires the computation of VaR
- CVaR-minimization might be **even harder** than VaR-minimization problem?

Under quite reasonable modeling assumptions, the opposite is true because that CVaR is coherent!

CVaR **more preferred** than VaR in risk optimization

- VaR is difficult to optimize numerically when losses are not normally distributed
- CVaR_α **preserves convexity** and can be expressed as a minimization formulation
- $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$ always holds, a **conservative convex upper-bound approximation**
- Able to control the **expected tail risk**



- Now consider the problem

$$\min \{ \mathcal{R}[l(x, \xi)] : x \in \mathcal{A} \}$$

Theorem:

Assume $\mathcal{R}(\cdot)$ is a **convex and monotone** risk function and $l(x, \xi^s)$ is convex for $s = 1, \dots, S$. Then $\mathcal{R}[l(x, \xi)]$ is a convex function of x

- Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^S$ be defined by $h_s(x) = l(x, \xi^s)$
- $h(\cdot)$ is a convex function. $\mathcal{R}(\cdot)$ is convex and increasing function.
- $\Rightarrow \mathcal{R}(h(x))$ is convex.



- Recall $l(x, \xi)$ represents **a random loss function**
- Consider the auxiliary function

$$\begin{aligned} F_\alpha(x, \gamma) &= \gamma + \frac{1}{1 - \alpha} \mathbb{E} \{ [l(x, \xi) - \gamma]_+ \} \\ &= \gamma + \frac{1}{1 - \alpha} \int_{\xi \in \Xi} [l(x, \xi) - \gamma]_+ p(\xi) d\xi \end{aligned}$$

Theorem

1. For any fixed x , the function $\gamma \mapsto F_\alpha(x, \gamma)$ is convex.
2. $\mathbf{VaR}_\alpha(x)$ is a minimizer of the problem $\min_\gamma F_\alpha(x, \gamma)$.
3. $F_\alpha(x, \mathbf{VaR}_\alpha(x)) = \mathbf{CVaR}_\alpha(x)$.



Proof: i) Since $[l(x, \xi) - \gamma]_+$ is a convex function in γ , it is true that this convexity **is preserved by integrals and linear combination** (Page 79, Boyd, S. and Vandenberghe, L., 2004. Convex optimization. Cambridge university press)

ii) Since $\min_{\gamma} F_{\alpha}(x, \gamma)$ is convex based on theorem 1(i), the KKT conditions are sufficient for optimality, i.e., we only need to check that $F_{\alpha}(x, \gamma)$ is stationary at $\gamma = \text{VaR}_{\alpha}(x)$

$$\frac{\partial F_{\alpha}(x, \gamma)}{\partial \gamma} \Big|_{\gamma = \text{VaR}_{\alpha}(x)} = 0$$



ii) Based on the following Proposition (Shapiro & Wardi 1994):

$E'[f(x)] = E[f'(x)]$ when the function $f(x)$ is convex w.p.1

we have

$$\begin{aligned}\frac{\partial F_\alpha(x, \gamma)}{\partial \gamma} &= 1 + \frac{1}{1 - \alpha} \int_{\xi \in \Xi} [-\mathbb{I}_{l(x, \xi) \geq \gamma}] p(\xi) d\xi \\ &= 1 - \frac{P[l(x, \xi) \geq \gamma]}{1 - \alpha}\end{aligned}$$

$$P[l(x, \xi) \geq \text{VaR}_\alpha(x)] = 1 - \alpha$$

$$\left. \frac{\partial F_\alpha(x, \gamma)}{\partial \gamma} \right|_{\gamma = \text{VaR}_\alpha(x)} = 0$$

Shapiro, A., and Y. Wardi. "Nondifferentiability of the steady-state function in discrete event dynamic systems." *IEEE transactions on Automatic Control* 39, no. 8 (1994): 1707-1711.

iii) We have

$$\begin{aligned} F_{\alpha}(x, \text{VaR}_{\alpha}(x)) &= \text{VaR}_{\alpha}(x) + \frac{1}{1-\alpha} \int_{\xi \in \Xi} [l(x, \xi) - \text{VaR}_{\alpha}(x)]_+ p(\xi) d\xi \\ &= \text{VaR}_{\alpha}(x) + \boxed{\frac{1}{1-\alpha} \int_{\{\xi \in \Xi: l(x, \xi) \geq \text{VaR}_{\alpha}(x)\}} l(x, \xi) p(\xi) d\xi} \\ &\quad - \frac{1}{1-\alpha} \int_{\{\xi \in \Xi: l(x, \xi) \geq \text{VaR}_{\alpha}(x)\}} \text{VaR}_{\alpha}(x) p(\xi) d\xi \\ &= \text{VaR}_{\alpha}(x) + \text{CVaR}_{\alpha}(x) \\ &\quad - \frac{\text{VaR}_{\alpha}(x)}{1-\alpha} \boxed{\int_{\{\xi \in \Xi: l(x, \xi) \geq \text{VaR}_{\alpha}(x)\}} p(\xi) d\xi} \\ &= \text{VaR}_{\alpha}(x) + \text{CVaR}_{\alpha}(x) - \text{VaR}_{\alpha}(x) \end{aligned}$$



The theorems imply

$$\begin{aligned}\min_{x \in \mathcal{X}} \text{CVaR}_\alpha[l(x, \xi)] &\Leftrightarrow \min_{x, \gamma \in \mathcal{X} \times \mathbb{R}} F_\alpha[l(x, \xi), \gamma] \\ &\Leftrightarrow \min_{x, \gamma \in \mathcal{X} \times \mathbb{R}} \gamma + \frac{1}{1 - \alpha} \mathbb{E} \{[l(x, \xi) - \gamma]_+\}\end{aligned}$$

- If the loss function $l(x, \xi)$ is convex in x , then $F_\alpha(x, \gamma)$ is convex in x and γ , the problem is **convex optimization** that can be well solved
- If (x^*, γ^*) minimizes $F_\alpha(x, \gamma)$ over $\mathcal{X} \times \mathbb{R}$, then $F_\alpha(x^*, \gamma^*)$ is the optimal value of the CVaR minimization problem.



- If CVaR is in the constraint

$$\mathbf{CVaR}_\alpha[l(x, \xi)] \leq L \Leftrightarrow \exists \gamma \in \mathbb{R}, F_\alpha[l(x, \xi), \gamma] \leq L$$

- Thus,

$$\begin{cases} \min_{x \in \mathcal{X}} & f_0(x) \\ \text{s.t.} & \mathbf{CVaR}_\alpha[l(x, \xi)] \leq L \end{cases} \Leftrightarrow \begin{cases} \min_{x, \gamma \in \mathcal{X} \times \mathbb{R}} & f_0(x) \\ \text{s.t.} & F_\alpha[l(x, \xi), \gamma] \leq L \end{cases}$$

- Convex optimization when $l(x, \xi)$ is convex in x

- In the **discrete case** (e.g., sampling)

$$F_{\alpha}(x, \gamma) = \gamma + \frac{1}{1 - \alpha} \sum_{k=1}^N p_k [l(x, \xi^k) - \gamma]_+$$

- Replace $[l(x, \xi^k) - \gamma]_+$ by additional variables z^k ,

$$\begin{aligned} \min_{x \in \mathcal{X}} \mathbf{CVaR}_{\alpha}[l(x, \xi)] &\Leftrightarrow \min_{x, \gamma \in \mathcal{X} \times \mathbb{R}} \gamma + \frac{1}{1 - \alpha} \sum_{k=1}^N p_k z^k \\ &\text{s.t. } z^k \geq 0, \forall k \\ &\quad z^k \geq l(x, \xi^k) - \gamma, \forall k \end{aligned}$$

- When $l(x, \xi)$ affine in x , the CVaR minimization is an LP
- When $l(x, \xi)$ convex in x , convex optimization



- Optimal dike construction

$$\begin{aligned} \min_x \quad & 0.9x + \frac{1}{N} \sum_{k=1}^N [b_k (H_k - x)_+] \\ \text{s.t.} \quad & 0 \leq x \leq 3 \end{aligned}$$

$c = 0.9$, $H \sim \text{Gamma}(8, 4.5)$ and $b \sim \text{Lognorm}(0.7, 0.16^2)$

$$\begin{aligned} \min_{x, \gamma \in [0, 3] \times \mathbb{R}} \quad & \gamma + \frac{1}{(1 - \alpha)N} \sum_{k=1}^N z^k \\ \text{s.t.} \quad & z^k \geq 0, \forall k \\ & h^k \geq H_k - x, h_k \geq 0, \forall k \\ & z^k \geq 0.9x + b_k h_k - \gamma, \forall k \end{aligned}$$

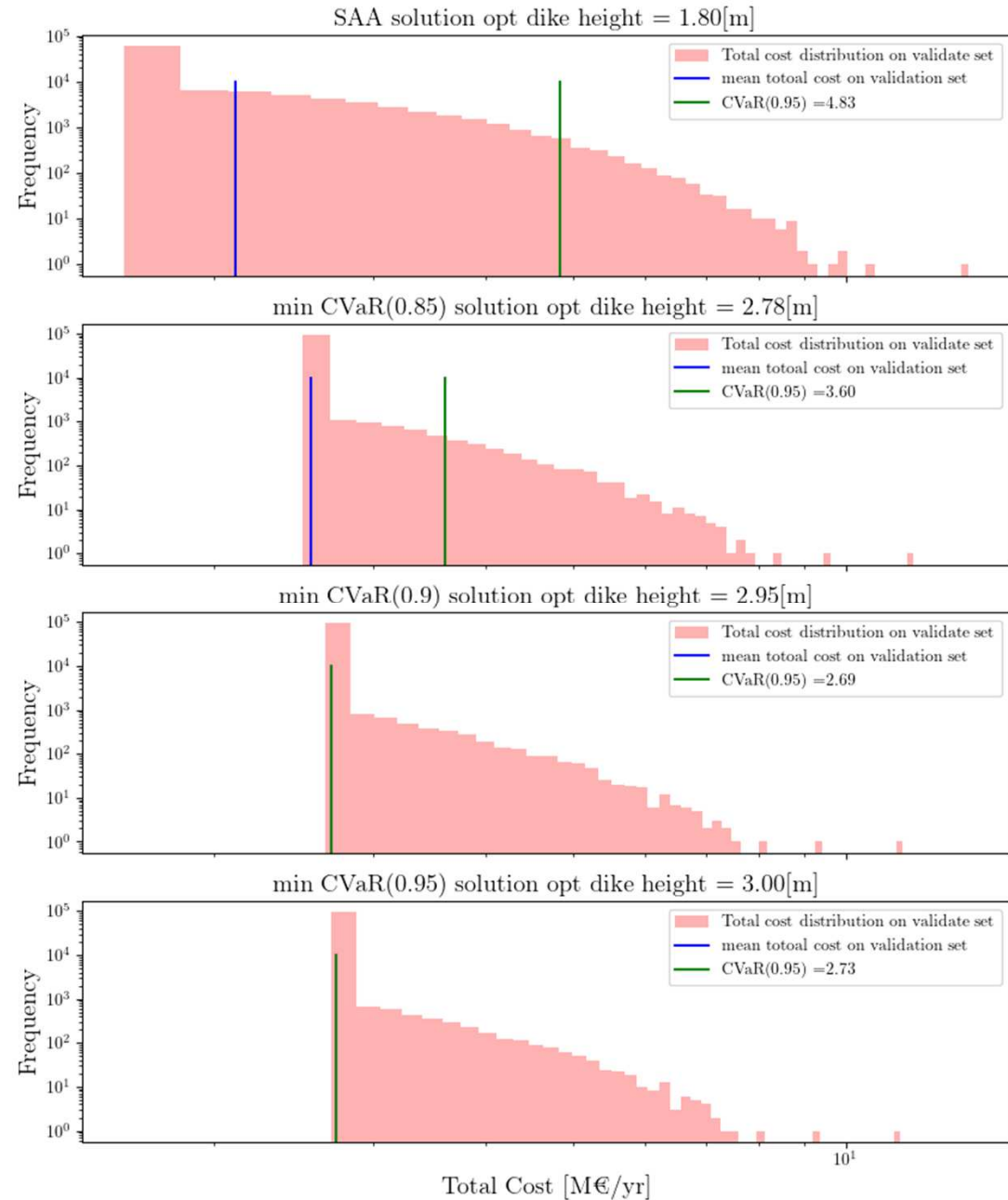


CVaR optimization example



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$(N = 2000)$	x [m]
SAA	1.80
CVaR _{0.85}	2.78
CVaR _{0.9}	2.95
CVaR _{0.95}	3.00





When to use what?



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Measures	Best for	Drawbacks
Mean	• Risk-neutral • Uncertainty averages out over time	• Ignores variability • Can lead to high volatility or catastrophic failures
MV	• Symmetric uncertainty, risk can be captured by variance	• Fails for asymmetric & heavy-tailed distributions • Doesn't distinguish upside & downside risk
VaR	• Regulatory risk limits • Need to control the probability of extreme loss	• Nonconvex, hard to optimize • Doesn't account for the severity of tail losses
CVaR	• Minimize extreme downside risk • Heavy-tailed & asymmetric uncertainty	• Sensitive to tail behavior (errors in estimating the tail distribution)



Exercise



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- Implement the risk-averse dike construction model



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